



illuminati - 2020

Advanced Physics Assignment-3

Vectors, Kinematics, 2D-Motion, Dynamics of A particle,
Energy and Momentum, Rotational Motion, Gravitation

SECTION 1

SINGLE CORRECT ANSWER TYPE

1.(B) $I_T = m(v - 0); I_T = m(v_0 \cos \theta - v)$

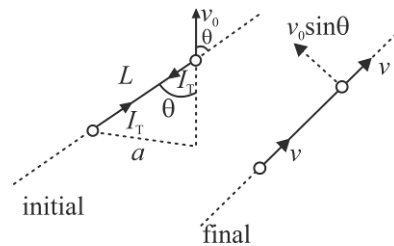
$$mv = mv_0 \cos \theta - mv; \quad 2mv = mv_0 \cos \theta$$

$$v = \frac{v_0}{2} \cos \theta; \quad \Delta K = KE_i - KE_f$$

$$= \frac{1}{2}mv_0^2 - \frac{1}{2}(2m)v^2 - \frac{1}{2}mv_0^2 \sin^2 \theta$$

$$= \frac{1}{2}mv_0^2 - \frac{1}{2}(2m)\frac{v_0^2}{4} \frac{(L^2 - a^2)}{L^2} - \frac{1}{2}mv_0^2 \frac{a^2}{L^2}$$

$$= \frac{1}{2}mv_0^2 \left(1 - \frac{(L^2 - a^2)}{2L^2} - \frac{a^2}{L^2} \right) = \frac{1}{2}mv_0^2 \left(\frac{L^2 - a^2}{2L^2} \right) = \frac{mv_0^2}{4L^2} (L^2 - a^2)$$



2.(C) $(\sum F_x)_{\text{System}} = 0; \quad (a_{\text{com}})_x = 0$

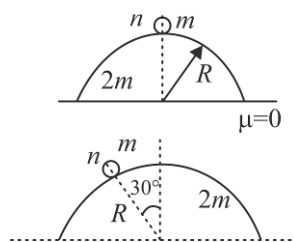
$$(V_{\text{com}})_x = 0 \quad \therefore \quad (\Delta x_{\text{com}})_x = 0$$

$$m_{ss} x_{ss} + m_{bs} X_{bs} = 0$$

$$m_{ss} (x_{ss/bs} + X_{bs/G}) + m_{bs} X_{bs/G} = 0$$

$$m((-R+r) \sin 60^\circ + X) + 2m(X) = 0$$

$$3mX = m(R+r) \frac{1}{2}; \quad X = \frac{(R+r)}{6}$$

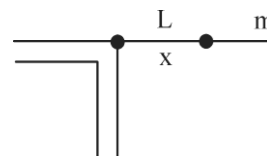


3.(C) $\tau = I\alpha; \quad mg(L) + mg(x) = (mL^2 + mx^2)\alpha$

$$\alpha = g \frac{(L+x)}{L^2 + x^2}$$

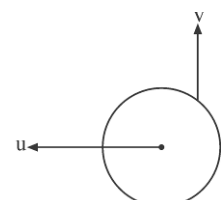
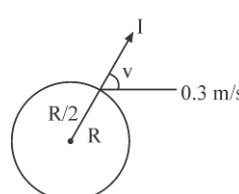
$$\frac{d\alpha}{dx} = 0 \Rightarrow (L^2 + x^2)(1) - (L+x)(2x) = 0$$

$$L^2 + x^2 - 2Lx - 2x^2 = 0 \quad \therefore \quad x = (\sqrt{2} - 1)L$$



4.(A) $\cos \theta = \sqrt{3}/2 \sin \theta \frac{1}{2}; \quad I \cos \theta = m(0.3)$

$$I \sin \theta = m(v); \quad \tan \theta = \frac{v}{0.3}$$



$$v = \frac{3}{10} \times \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{10} \text{ m/s}; \quad v_{sep} = v_{app}$$

$$\frac{v\left(\frac{1}{2}\right) + u\left(\frac{\sqrt{3}}{2}\right)}{3\left(\frac{\sqrt{3}}{2}\right) - u\left(\frac{\sqrt{3}}{2}\right)} = 1; \quad \frac{\sqrt{3}}{20} + \frac{\sqrt{3}u}{2} = \frac{3\sqrt{3}}{20} - \frac{\sqrt{3}}{2}u; \quad \sqrt{3}u = \frac{2\sqrt{3}}{20}; \quad u = \frac{1}{10} \text{ m/s}$$

$$5.(B) \quad \frac{2m(\ell/2) + 2m\left(\ell/2 + \frac{\ell}{10}\right) + 2m\left(\ell/2 + \frac{2\ell}{10}\right)}{5m}; \quad \frac{\ell + \frac{12\ell}{10} + \frac{7\ell}{10}}{5} = \frac{29\ell}{50}$$

$$6.(A) \quad \frac{1m}{2}(500)^2 \times 1.5 = KE_f$$

$$m(v_0) - 2m(v_1) = 3m(500)$$

$$v_0 = 2v_1 + 3(500)$$

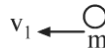
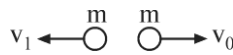
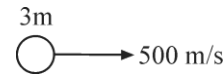
$$\frac{1}{2} \times 3m(500)^2 \times 1.5 = \frac{1}{2}m(v_0^2) + \frac{2.1}{2}m(v_1^2)$$

$$3 \times 500^2 \times 1.5 = (2v_1 + 1500)^2 + 2v_1^2$$

$$3 \times 500^2 \times 1.5 = 4v_1^2 + 2v_1^2 + (1500)^2 + 6000v_1$$

$$6v_1^2 + 6000v_1 + 1500^2 - 3 \times 1.5 \times 500^2 = 0;$$

$$v_1 = -250 \text{ m/s}; \quad v_1 = -750 \text{ m/s};$$



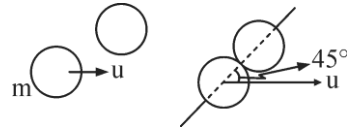
$$6v_1^2 + 6000v_1 + 500^2(9 - 4.5) = 0$$

$$v_0 = 2(-250) + 1500 = 1000 \text{ m/s}$$

$$7.(B) \quad mu \cos \theta = 2mv$$

$$v = \frac{u \cos \theta}{2}; \quad PE / KE = \frac{\frac{1}{2}m(u \cos \theta)^2 - \frac{1}{2}m\left(\frac{u \cos \theta}{2}\right)^2 \times 2}{\frac{1}{2}mu^2}$$

$$= \frac{\left(\frac{u^2}{\sqrt{2}}\right)^2 - \left(\frac{u}{2\sqrt{2}}\right)^2 \times 2}{u^2} = \frac{1}{2} - \frac{1}{4 \times 2} \times 2 = \frac{1}{4}$$



$$8.(A) \quad \downarrow \sqrt{2gh} \uparrow 1/n \sqrt{2gh} \downarrow \sqrt{2gh}/n \uparrow \sqrt{2gh}/n^2$$

$$= m(\sqrt{2gh}) \left[(1 + 1/n) + (1/n + 1/n^2) + (1/n^2 + 1/n^3) - \dots \right] = m\sqrt{2gh} (1 + n) \left(\frac{1}{n-1} \right)$$

$$9.(C) \quad KE = \frac{1}{2} \left(\frac{2}{5} mr^2 \right) \frac{v^2}{r^2} + \frac{1}{2} \left(\frac{2}{5} mr^2 + mR^2 \right) \frac{v^2}{R^2}$$

$$= \frac{1}{5} mv^2 + \frac{1}{5} m \frac{v^2 r^2}{R^2} + \frac{1}{2} mv^2; \quad = \frac{7}{10} mv^2 \left(1 + \frac{2}{7} \frac{r^2}{R^2} \right)$$

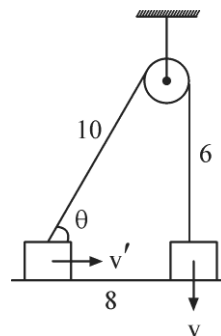
$$10.(A) \quad W_f = -\mu mg \cdot R; \quad W_g = -mgR$$

$$W_f + W_g + W_T = 0; \quad W_T = mgR(\mu + 1)$$

$$11.(B) \quad \cos \theta = \frac{8}{10}$$

$$\text{Cons. of energy } v^1 = \frac{10}{8} v$$

$$mg(5) = \frac{1}{2} mv^2 + \frac{1}{2} mv'^2; \quad v = \frac{40}{\sqrt{41}} \text{ m/s}$$



12.(C) $df = \mu \left(\frac{m}{l} dx \right) g(x); \quad f = -\frac{mg l}{9}$

13.(A) $S = \frac{\ell}{\cos \theta}; \quad a = g \sin \theta - k g \cos \theta$

$$\frac{\ell}{\cos \theta} = \frac{1}{2} (g \sin \theta - k g \cos \theta) t^2$$

$$t^2 = \frac{2\ell}{g \cos \theta (\sin \theta - k \cos \theta)} = \frac{2}{g} \frac{(\sec \theta)}{(\sin \theta - k \cos \theta)}$$

$$\frac{d}{d\theta} (\sin \theta \cos \theta - k \cos^2 \theta) = 0; \quad \sin \theta (-\sin \theta) + \cos^2 \theta - 2k \cos \theta (-\sin \theta) = 0$$

$$\cos 2\theta + k \sin 2\theta = 0; \quad \tan 2\theta = -\frac{1}{k}$$

$$\theta = \frac{1}{2} \tan^{-1} \left(-\frac{1}{k} \right)$$

14.(B) Using conservation of angular momentum about O.

$$m V_P r_P = m V_A r_A = m V_0 r_0 \cos \theta; \quad V_A r_A = V_P r_P = \frac{3v_0 r_0}{5}$$

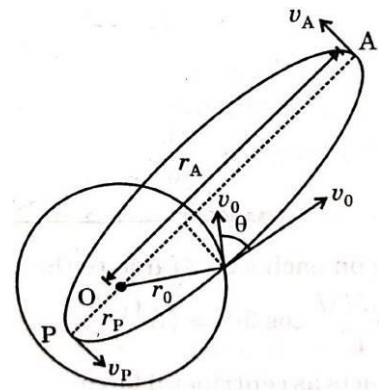
Using conservation of energy.

$$\frac{1}{2} m V_A^2 + \frac{-GMm}{r_A} = \frac{-GMm}{r_0} + \frac{1}{2} m V_0^2$$

$$\Rightarrow \frac{9V_0^2 r_0^3}{50r_A^2} - \frac{V_0^2 r_0}{r_A} + \frac{V_0^2}{2} = 0 \quad \left[\text{Let } \frac{r_0}{r_A} = x \right]$$

$$\Rightarrow 9x^2 - 50x + 25 = 0 \Rightarrow x = 5 \text{ or } \left(\frac{5}{9} \right)$$

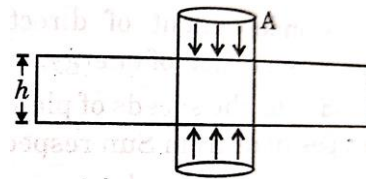
$$\Rightarrow r_A = \frac{9r_0}{5}; \quad r_P = \frac{r_0}{5} \Rightarrow \frac{V_P}{V_A} = \frac{r_A}{r_P} = 9$$



15.(A) Gauss law for gravitation

$$\int \vec{g} \cdot d\vec{s} = -m_{in} \cdot 4\pi G \Rightarrow g = \frac{GM}{R^2}$$

$$2 \times \frac{GM}{R^2} \times A = \frac{M}{\frac{4}{3}rR^3} (h \times A) \times 4\pi G \Rightarrow h = \frac{2R}{3}$$



SECTION 2

LINK-COMPREHENSION TYPE

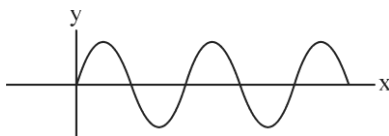
16.(B) $x = 2t; \quad y = 10 \sin(2t)$

$$v_x = 2 \quad a_x = 0$$

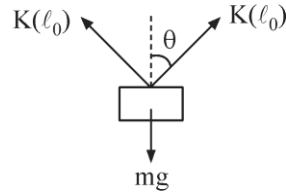
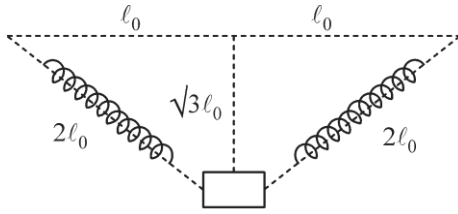
$$v_y = 20 \cos 2t; \quad a_y = -40 \sin 2t; \quad a \propto y$$

17.(C) $v = \sqrt{4 + (b \cos 2t)^2} = \sqrt{4 + 400} = \sqrt{404} = 2\sqrt{101}$

18.(D) $y = 10 \sin x$



19.(C)



$$2k l_0 \cos \theta = mg$$

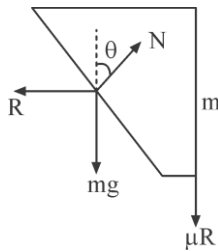
$$2kl_0 \frac{\sqrt{3}}{2} = mg ; \quad m = \frac{kl_0 \sqrt{3}}{g}$$

20.(B) Energy constant

$$mg \sqrt{3}l_0 = \frac{1}{2}kl_0^2 \times 2 + \frac{1}{2}mv^2 ; \quad kl_0^2(3) - kl_0^2 = \frac{mv^2}{2}$$

$$v^2 = 2 \times 2 \frac{kl_0^2}{m} = 4 \times \frac{g}{10\sqrt{3}} v_0^2 ; \quad v = \sqrt{\frac{4l_0 g}{\sqrt{3}}}$$

21.(B) 22.(C)

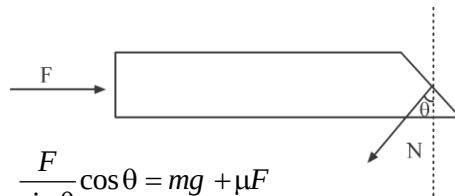


$$\sin \theta = \frac{5}{3}$$

$$\cos \theta = \frac{4}{5}$$

$$N \cos \theta = mg + \mu R$$

$$N \sin \theta = R$$



$$F = N \sin \theta$$

$$F = R$$

$$\frac{F}{\sin \theta} \cos \theta = mg + \mu F$$

$$F \left(\frac{4}{3} - \frac{2}{3} \right) = mg ; \quad F = \frac{3}{2} mg$$

(B) $F - N \sin \theta = 2ma_1 \rightarrow a_1 \uparrow a_2$

$$N \cos \theta - mg - \frac{1}{3}(N \sin \theta) = ma_2$$

$$a_2 \frac{4}{5} = a_1 \frac{3}{5} ; \quad a_1 = \frac{4a_2}{3}$$

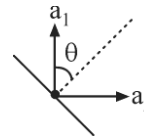
$$\left(\frac{F - 2ma_1}{\sin \theta} \right) \cos \theta - mg - \frac{1}{3} \left(\frac{F - 2ma_1}{\sin \theta} \right) \sin \theta = \frac{m3a_1}{4}$$

$$(F - 2ma_1) \frac{4}{3} - mg - \frac{1}{3} (F - 2ma_1) = \frac{3}{4} ma_1$$

$$F - mg = \frac{11}{4} ma_1 ; \quad \cancel{\frac{mg}{2}} = \frac{11}{4} \cancel{a_1}$$

$$\frac{mg}{2} = \frac{11}{4} ma_1$$

$$a_1 = \frac{2g}{11} ; \quad a_2 = \frac{3g}{22}$$



32.(D) Work done by all force is equal $K_f - K_i = \frac{1}{2} \left(\frac{m}{2} \right) v^2 = \frac{mR^2\omega_0^2}{9}$

33.(C) $\frac{2G\lambda}{r} \times m = \frac{mv^2}{r}$

Where $\lambda = \text{mass per unit length} = \rho \times A = \rho \times \pi R^2 \Rightarrow \frac{2G \times \rho \times \pi R^2}{r} = \frac{v^2}{r} \Rightarrow v = R\sqrt{2\pi G\rho}$

34.(B) $\int_R^{3R} \frac{2G\lambda_m}{r} dr = \frac{1}{2}mv^2 \Rightarrow v = 2R\sqrt{\pi\rho g \ln 3}$

35.(D) $T = \frac{2\pi d}{v} \Rightarrow d = RT\sqrt{\frac{G\rho}{2\pi}}$

SECTION 3

MULTIPLE CORRECT ANSWER TYPE

36.(BC) at any x

$$\frac{1}{2}k(x_0^2) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2; \quad \frac{k}{m}(x_0^2 - x^2) = v^2$$

$$v = \sqrt{\frac{k}{m}(x_0^2 - x^2)}; \quad F = kx$$

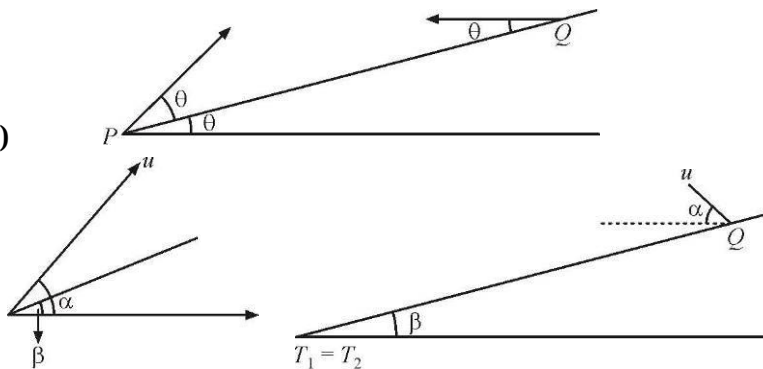
Mag. of power $F_v = kx\sqrt{\frac{k}{m}(x_0^2 - x^2)}$ (sing depends on distinct)

$$= kx\sqrt{\frac{k}{m}(x_0^2 - x^2)}$$

For max. power $dp/dx = 0; \quad kx\sqrt{\frac{k}{m} \frac{1}{2} \frac{(-2x)}{\sqrt{x_0^2 - x^2}}} + \sqrt{\frac{k}{m}(x_0^2 - x^2)}k = 0; \quad x_0^2 = 2x^2 \quad [x = \pm x_0 / \sqrt{2}]$

$$P\left(\frac{x_0}{\sqrt{2}}\right) = \frac{kx_0}{\sqrt{2}} \cdot \sqrt{\frac{k}{m}\left(x_0^2 - \frac{x_0^2}{2}\right)} = \frac{km_0}{2} \sqrt{\frac{k}{m}} x = \frac{k}{2} \sqrt{\frac{k}{m}} x_0^2$$

37.(BD)



Further acceleration of both the particles is g downwards. Therefore, relative acceleration between the two is zero or relative motion between the two is uniform. Now relative velocity of P with respect to Q is towards PQ . Therefore, collision will take place between the two in mid-air.

38.(AD) In the upward journey resistance due to air will be in downward direction. While in downward motion resistance will be upwards. Therefore, t_1 will decrease, while t_2 will increase. Hence, $T = t_1 + t_2$ may increase or decrease. H will decrease. In horizontal direction air resistance is opposite to the motion. Therefore, R_1 will decrease while R_2 may decrease or increase.

39.(AD) $80g - 2T = 80a_1$; $4T - 20g = 20a_2$; $a_2 = \frac{a_1}{2}$

$$160g - 4T = 160a_1$$

$$4T - 20g = 20a_2$$

$$140 = 340a_2$$

$$a_2 = \frac{7g}{17}; \quad a_1 = \frac{14g}{17}$$

40.(AC) $U_{\min}$ is at $\sin(4\pi x) = -1$

$$U_{\min} = 20 - 5 = 15 J$$

$$K_{\max} = E - U_{\min} = 20 - 15 = 5 J$$

$$4\pi x = \frac{3}{2}\pi, \frac{7}{2}\pi; x = \frac{3}{8}, \frac{7}{8}, \dots$$

41.(ABCD) $h = s \sin \theta$

$$\frac{1}{2}mv^2 + mg(H - h) = mgH; \quad \frac{1}{2}mv^2 - mgh = \text{zero}$$

$$v^2 - 2gh = \text{zero}; v^2 - 2gs(\sin \theta) = \text{zero}$$

But $\sin \theta = \frac{H}{P} = \frac{(H - h)}{(p - s)}$ So, $v^2 - 2gs \frac{(H - h)}{(p - s)} = \text{zero}$ and $v^2 - 2gs \frac{H}{p} = \text{zero}$

42.(AB) $\omega = \frac{V_{\text{relative}}}{\text{perpendicular distance}}; \quad \omega = \frac{V_1 - V_2}{l} = \frac{3t - t}{1}$

$$\omega = 2t; \quad \alpha = \frac{d\omega}{dt} = 2 \text{ rad/sec}^2$$

$$Ma_{\text{com}} = (T_1 + T_2) - Mg \quad \dots(i)$$

$$V_{\text{com}} \text{ w.r.t. } O' = \omega \times \frac{1}{2}; \quad V_{\text{com}} \text{ w.r.t. } O' = t$$

$$V_{\text{com}} \text{ w.r.t. } O' = V_{\text{com}} - V_{O'}; \quad t = V_{\text{com}} - t$$

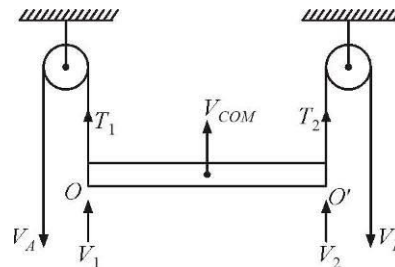
$$\Rightarrow V_{\text{com}} = 2t; \quad a_{\text{com}} = \frac{dv}{dt} = 2 \text{ m/s}$$

$$M(a + g) = T_1 + T_2 \quad [\text{From (i)}]$$

$$\Rightarrow T_1 + T_2 = 60 \quad \dots(ii)$$

$$I\alpha = \tau_{\text{com}}; \quad \frac{Ml^2}{12} \times 2 = (T_1 - T_2)R$$

$$\Rightarrow T_1 - T_2 = \frac{5}{3} \quad \dots(iii) \quad \text{Adding (ii) and (iii)} \quad T_1 = \frac{185}{6} N \text{ and } T_2 = \frac{175}{6} N$$



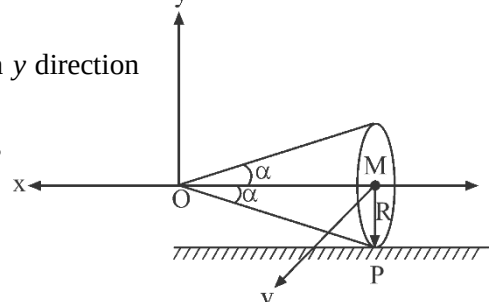
*43.(AB) **Method 1 x acceleration**

If we see carefully then x direction is rotating which y direction is not rotating.

Angular velocity of disc with respect to centre M is

$$\vec{\omega}_{D-M} = \frac{V}{R} \hat{i} \quad (i)$$

Angular velocity of centre M w.r.t origin



$$\vec{\omega}_{M-O} = \frac{v}{OM} \hat{j}$$

$$OM = \cot \alpha$$

$$\vec{\omega}_{M-O} = \frac{V}{R} \tan \alpha \hat{j} \quad (ii)$$

Since angular velocity is vector quantity it follow vector addition law.

$$\vec{\omega}_{D-O} = \frac{V}{R} [\hat{i} + \tan \alpha \hat{j}]$$

$$|\vec{\omega}_{D-O}| = \frac{V}{R} \sqrt{1 + \tan^2 \alpha} = \frac{V}{R \cos \alpha}$$

Again

Angular acceleration (β) is

$$\vec{\beta} = \frac{d\vec{\omega}}{dt} \quad ; \quad \vec{\beta} = \frac{V}{R} \left[\frac{d\hat{i}}{dt} + \tan \alpha \frac{d\hat{j}}{dt} \right]$$

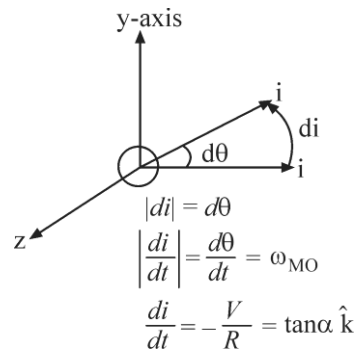
Add since y-direction is constant

$$\frac{dy}{dt} = 0$$

$$\text{But } \frac{d\hat{i}}{dt}$$

$$\vec{\beta} = \left(\frac{V}{R} \right) \left(\frac{V}{R} \right) \tan \alpha \hat{k}$$

$$\beta = -\frac{V^2}{R^2} \tan \alpha \hat{k} \quad ; \quad |\vec{\beta}| = \frac{V^2}{R^2} \tan \alpha$$



44.(BCD) Let a_2 be acceleration of 8kg block

$$8g - 8T = 8a_2 \quad \dots(i)$$

Let a_2 be acceleration of 8kg block

$$\text{From (i)} \quad 80 = 8T; \quad T = 10 \text{ N}$$

$$\text{From (ii)} \quad f = 10 \text{ N}$$

45.(ABC) Normal force by block (m_2)

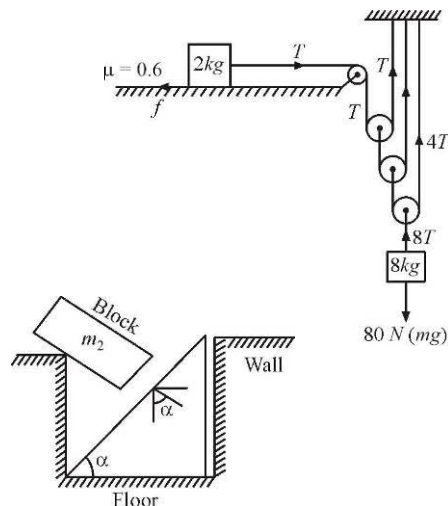
$$N = m_2 g \cos \alpha \quad [\text{by normal constant}]$$

$$\text{Normal force by wall } N = (m_2 g \cos \alpha) \sin \alpha$$

$$[\text{By normal constant}] = m_2 g \sin \alpha \cos \alpha$$

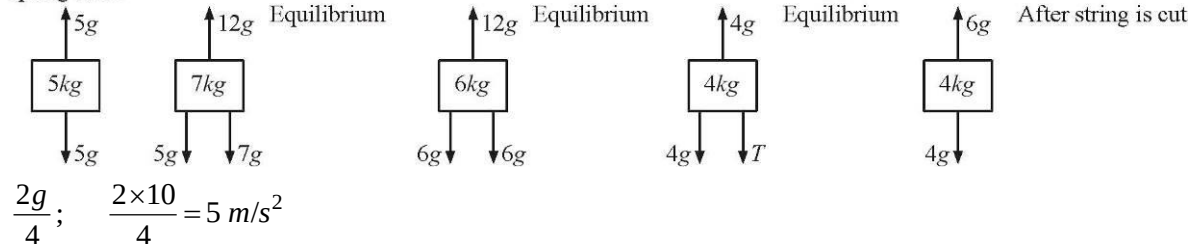
Normal force by floor

$$N = (m_2 g \cos \alpha) \cos \alpha + m_1 g = m_2 g \cos^2 \alpha + m_1 g$$



46.(ABD) Option (A) and (B) are correct

Spring force



47.(ABC) Let acceleration of block A be a_A and that of block B be a_B and velocities be v_A and v_B respectively

By string constant

$$3a_A = 2a_B$$

$$a_B = 6 \text{ m/s}^2 \quad (\text{Given})$$

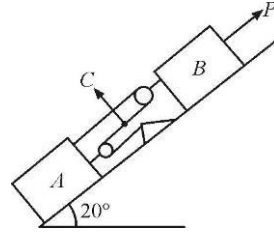
$$a_A = 4 \text{ m/s}^2$$

Also by string constant

$$3v_A = 2v_B$$

$$v_B = 3 \text{ m/s} \quad (\text{Given})$$

$$v_A = 2 \text{ m/s}$$



As one end of the string is attached to a fixed point it cannot provide any loose string for motion.

For pulley to have 2 m/s velocity

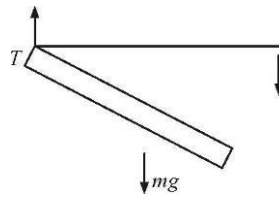
$$(v_A) = \frac{\text{velocity of middle string} + 0}{2}; \quad 4 = v_{\text{middle string}}$$

So point C will have 4 m/s velocity w.r.t. the ground.

$$48.(\text{CD}) \quad mg - T = ma; \quad T \frac{l}{2} \cos \theta = \frac{ml^2}{12} \alpha$$

$$\text{By constraints; } a = \frac{\alpha l}{2} \cos \theta$$

$$\text{Thus; } T = \frac{mg}{1 + \cos^2 \theta} \quad \text{and} \quad a_A = \frac{3g \sin \theta \cos \theta}{1 + 3 \cos^2 \theta}$$



49.(A) Let the mass of the ring be m , let its radius be R , and let the angular velocity of the system when the ring is at its lowest position be ω

Let the force exerted by any one rod be T

Due to symmetry at the instant when the rod is at the lowest position, the force exerted by the other rod is also T

Since the centre of mass of the ring is moving in circular path of radius $\frac{2R}{\pi}$,

$$2T \cos 30^\circ - mg = m\omega^2 \left(\frac{2R}{\pi} \right)$$

$$T = 144 \text{ N}$$

50.(B) Ω (Angular vel. About z-axis) $\alpha = 30^\circ \Rightarrow \beta = 45^\circ$

For Rolling :

$$\sin \alpha \cdot \Omega = \sin \beta \cdot \omega \Rightarrow \sin 30^\circ \cdot \Omega = \sin 45^\circ \cdot \omega \Rightarrow \Omega = \sqrt{2} \omega$$

$$L_{CM} = \frac{(ml^2 \sin^2 \beta) \omega}{2} = \frac{ml^2 \omega}{2} \cdot \frac{1}{2} = \frac{ml^2 \omega}{4}$$

$$\tau_0 = L_{CM} \sin 75^\circ \cdot \Omega = \frac{ml^2 \omega}{4} \cdot \frac{\sqrt{3}+1}{2\sqrt{2}} \cdot \sqrt{2} \omega = \frac{ml^2 \omega^2}{8} (\sqrt{3}+1)$$

$$L_{CM,z} = L_{CM} \cos 75^\circ = \frac{ml^2 \omega}{4} \cdot \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{ml^2 \omega}{8\sqrt{2}} (\sqrt{3}-1)$$

MJ / mole .

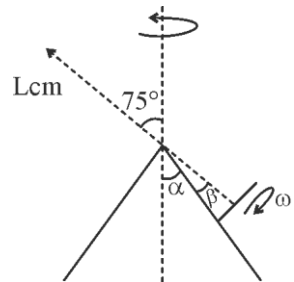
51.(ACD) When A is imparted a speed u , B starts with $u/2$.

u should be such that both A & B reach the topmost point of cir motion

$\Rightarrow$ using W/E the we write

Ki

$$\frac{1}{2} mu^2 + \frac{1}{2} m \left(\frac{u}{2} \right)^2 - mg(2n) - mg(4\ell) = 0^{kf}$$



$$\frac{1}{2} \left(\frac{5u^2}{4} \right) = 6gl ; \quad 5u^2 = 48gl ; \quad u = \sqrt{\frac{48gl}{5}} = 4\sqrt{\frac{3gl}{5}}$$

52.(AC)

Asteroid-I

Field $E_I = \frac{GM}{R^3} x$

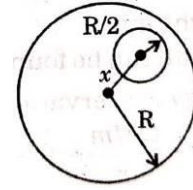
$$T_I = \frac{2\pi}{4} \sqrt{\frac{R^3}{GM}}$$

$$\frac{F_I}{F_{II}} = \frac{E_I}{E_{II}} = \frac{2x}{R}$$

Asteroid-II

Field $E_{II} = \frac{GM}{R^3} \left(\frac{R}{2} \right)$

$$T_{II} = 2\sqrt{\frac{R^2}{GM}}$$



53.(ACD) No external torque is acting in this case too.

So Kepler's law of areas is obeyed in this case too.

54.(AC)

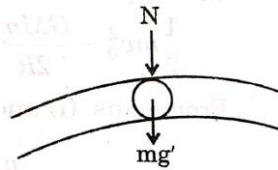
55.(AD) For contact force = 2 gravitational force

(i) $N + mg' = m\omega^2 r$

$$2mg' + mg' = m\omega^2 r$$

$$3g' = r\omega^2 ; \quad 3g \left(\frac{r}{R} \right) = r\omega^2$$

$$\omega = \sqrt{\frac{3g}{R}}$$



(ii) For $N = 0$, $g' = m\omega^2 r$

$$g \frac{r}{R} = m\omega^2 ; \quad \omega \text{ is independent of } r.$$

SECTION 4

COLUMN MATCHING TYPE

56.(C) Moment of inertia of the first disc about its centre, $I_1 = \frac{1}{2} mR^2$

Moment of inertia of the second disc about its centre, $I_2 = \frac{1}{2} (4m) \left(\frac{R}{2} \right)^2 = \frac{1}{2} mR^2 \Rightarrow \frac{I_1}{I_2} = 1$

From the force diagram, $mg \sin \theta - f = ma_{CM}$

About the centre of mass, $fr = I\alpha$

And, for no slipping, $a_{CM} = r\alpha$

Solving, we get $a_{CM} = \frac{g \sin \theta}{m + \frac{I}{r^2}}$

So, for first disc, $a_{CM} = a_1 = \frac{mg \sin \theta}{m + \frac{\left(\frac{1}{2} mR^2 \right)}{R^2}} = \frac{2}{3} g \sin \theta$

$$\text{And, for second disc, } a_{CM} = a_2 = \frac{(2m)g \sin \theta}{2m + \frac{\left(\frac{1}{4}mR^2\right)}{\left(\frac{R}{2}\right)^2}} = \frac{2}{3}g \sin \theta$$

$$\text{So, time taken to reach the bottom of the plane, } t = \sqrt{\frac{2\left(\frac{h}{\sin \theta}\right)}{a_{CM}}}$$

$$\text{Velocity of the centre at the bottom of the plane, } v = \sqrt{2a_{CM}\left(\frac{h}{\sin \theta}\right)}$$

$$\text{Angular momentum about the CM at the bottom of the plane, } L = I_{CM}\omega = \frac{I_{CM}v}{r}$$

$$\Rightarrow \frac{t_1}{t_2} = \sqrt{\frac{a_2}{a_1}} = 1, \quad \frac{v_1}{v_2} = \sqrt{\frac{a_1}{a_2}} = 1; \quad \frac{L_1}{L_2} = \left(\frac{I_1}{I_2}\right)\left(\frac{v_1}{v_2}\right)\left(\frac{r_2}{r_1}\right) = \frac{1}{2}$$

$$57.(A) \text{ Moment of inertia of the first disc about its centre, } I_1 = \frac{1}{2}m\left(R^2 + \left(\frac{R}{2}\right)^2\right) = \frac{5}{8}mR^2$$

$$\text{Moment of inertia of the second disc about its centre, } I_2 = \frac{1}{2}mR^2 \quad \Rightarrow \quad \frac{I_1}{I_2} = \frac{5}{4}$$

Writing the three equations and solving like the previous question, for first disc,

$$a_1 = \frac{mg \sin \theta}{m + \frac{\left(\frac{5}{8}mR^2\right)}{R^2}} = \frac{8}{13}g \sin \theta$$

$$\text{And, for second disc, } a_{CM} = a_2 = \frac{mg \sin \theta}{m + \frac{\left(\frac{1}{2}mR^2\right)}{R^2}} = \frac{2}{3}g \sin \theta$$

$$\text{As before, time taken to reach the bottom of the plane, } t = \sqrt{\frac{2\left(\frac{h}{\sin \theta}\right)}{a_{CM}}}$$

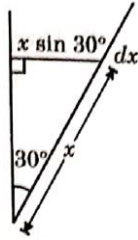
$$\text{Velocity of the centre at the bottom of the plane, } v = \sqrt{2a_{CM}\left(\frac{h}{\sin \theta}\right)}$$

$$\text{Angular momentum about the CM at the bottom of the plane, } L = I_{CM}\omega = \frac{I_{CM}v}{r}$$

$$\Rightarrow \frac{t_1}{t_2} = \sqrt{\frac{a_2}{a_1}} = \sqrt{\frac{13}{12}}, \quad \frac{v_1}{v_2} = \sqrt{\frac{a_1}{a_2}} = \sqrt{\frac{12}{13}}; \quad \frac{L_1}{L_2} = \left(\frac{I_1}{I_2}\right)\left(\frac{v_1}{v_2}\right)\left(\frac{r_2}{r_1}\right) = \sqrt{\frac{75}{52}}$$

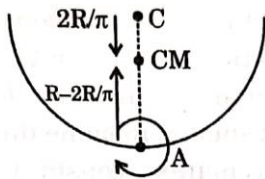
58. [A-Q] [B-P] [C-S] [D-R]

$$(A) \quad I = \int dm \left(\frac{x}{2} \right)^2 = \frac{m}{4l} \int x^2 dx = \frac{m}{4l} \left(\frac{x^3}{3} \right)^l = \frac{ml^2}{12}$$

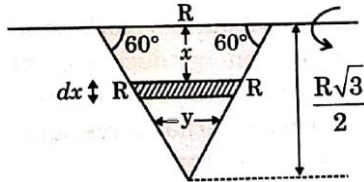


$$(B) \quad I_C = MR^2 = I_{cm} + M \left(\frac{2R}{\pi} \right)^2$$

$$I_A = I_{cm} + M \left(R - \frac{2R}{\pi} \right)^2 = \frac{8MR^2}{11}$$



$$(C) \quad \frac{\left(\frac{R\sqrt{3}}{2} - x \right)}{\frac{R\sqrt{3}}{2}} = \frac{R}{2} \quad \text{(An element is consider as shown)}$$



$$I = \int (dm)x^2 = \int (\sigma(dx)y)x^2 = \sigma \int \left(R - \frac{2x}{\sqrt{3}} \right) x^2 dx = \frac{M}{\left(\frac{\sqrt{3}R^2}{4} \right)} \left[R \frac{x^3}{3} - \frac{2}{\sqrt{3}} \frac{x^4}{4} \right]_0^{\frac{R\sqrt{3}}{2}} = \frac{MR^2}{8}$$

$$(D) \quad I = (I_{\text{bigger}})_0 - (I_{\text{smaller}})_0 = \frac{1}{2} M (2R)^2 - \left[\frac{1}{2} \frac{M}{4} R^2 + \frac{M}{2} R^2 \right] = 2MR^2 - \frac{3}{8} MR^2 = \frac{13}{8} MR^2$$

59. [A-S] [B-R] [C-S] [D-S]

(A) At maximum height $v = 0$ so from conservation of mechanical energy

$$\frac{1}{2} mu^2 - \frac{GMm}{R} = 0 - \frac{GMm}{R+h} = \frac{1}{2} (gR) - mgR = -\frac{mgR}{1 + \frac{h}{2}} \Rightarrow h = R$$

(B) Velocity of particle at a distance r from earth's centre is given by

$$\frac{GMm}{2R} - \frac{GMm}{R} = \frac{1}{2} mv^2 - \frac{GMm}{r}$$

$$\Rightarrow v = \sqrt{\frac{2GM}{r} - \frac{GM}{R}} \Rightarrow \frac{dr}{dt} = \sqrt{\frac{GM}{Rr} (2R-r)} \Rightarrow \int_R^{2R} \frac{\sqrt{r} dr}{\sqrt{2R-r}} = \int_0^T \sqrt{\frac{GM}{R}} dt$$

$$\text{Put } r = 2R \sin^2 \theta \Rightarrow dr = 4R \sin \theta \cos \theta d\theta$$

$$\Rightarrow \int_{\pi/4}^{\pi/2} \frac{\sqrt{2R} \sin \theta (4R \sin \theta \cos \theta d\theta)}{\sqrt{2R} \cos \theta} = \int_0^T \sqrt{\frac{GM}{R}} dt \Rightarrow T = \left(\frac{\pi}{2} + 1\right) \sqrt{\frac{R^3}{GM}}$$

(C) Acceleration due to gravity at $g_h = \frac{GM}{R^2 \left(1 + \frac{h}{R}\right)^2} = \frac{GM}{4R^2}$

(D) Orbital velocity $v_0 \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{2R}}$

60. [A-P] [B-R] [C-Q] [D-S]

$$v_e = \sqrt{\frac{2GM}{r}}$$

$v < v_e$: path is elliptical

$v = v_e$: path is parabolic

$v > v_e$: path is hyperbolic

61. [A-S] [B-T] [C-P]

Equation of trajectory: $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$

Remember that for the equation of trajectory, the point of projection is the origin.

Now, to find the condition for the ball to hit the wall, we should find the Y-coordinate, y_0 , of the ball when its X-coordinate becomes 20. And then apply the condition that $0 < y_0 < 10$.

$$\text{So, } y_0 = 20 \tan \theta - \frac{(10)(20)^2}{2(20)^2 \cos^2 \theta} = 20 \tan \theta - 5(1 + \tan^2 \theta)$$

Now, $0 < y_0 < 10$

Let us first consider $y_0 > 0$

$$\Rightarrow 20 \tan \theta - 5(1 + \tan^2 \theta) > 0 \Rightarrow \tan^2 \theta - 4 \tan \theta + 1 < 0$$

$$\Rightarrow 2 - \sqrt{3} < \tan \theta < 2 + \sqrt{3} \Rightarrow 15^\circ < \theta < 75^\circ$$

Now, let us consider $y_0 < 10$

$$\Rightarrow 20 \tan \theta - 5(1 + \tan^2 \theta) < 10$$

$$\Rightarrow \tan^2 \theta - 4 \tan \theta + 3 > 0$$

$$\Rightarrow \tan \theta < 1 \quad \text{OR} \quad \tan \theta > 3$$

$$\Rightarrow \theta < 45^\circ \quad \text{OR} \quad \theta > \tan^{-1}(3)$$

So, the condition for the ball to hit the wall is:

$$15^\circ < \theta < 45^\circ \quad \text{OR} \quad \tan^{-1}(3) < \theta < 75^\circ$$

Now, the ball falls between A and the wall if $y_0 < 0$

$$\Rightarrow 20 \tan \theta - 5(1 + \tan^2 \theta) < 0 \Rightarrow \tan^2 \theta - 4 \tan \theta + 1 > 0$$

$$\Rightarrow \tan \theta < 2 - \sqrt{3} \quad \text{OR} \quad \tan \theta > 2 + \sqrt{3}$$

$$\Rightarrow \theta < 15^\circ \quad \text{OR} \quad \theta > 75^\circ$$

And, finally the ball falls beyond the wall if $y_0 > 10$

$$\Rightarrow 20 \tan \theta - 5(1 + \tan^2 \theta) > 10 \Rightarrow \tan^2 \theta - 4 \tan \theta + 3 < 0$$

$$\Rightarrow 1 < \tan \theta < 3 \quad \text{OR} \quad \tan \theta > 3 \Rightarrow 45^\circ < \theta < \tan^{-1}(3)$$

62. [A – Q] [B – R] [C – S]

If the particle is able to reach the point where the net gravitational force due to both the planets is zero, it will be able to reach the surface of the second planet.

Let this point S be a distance x from A

$$\text{Then, } \frac{GMm}{x^2} = \frac{G(4M)m}{(4R-x)^2} \Rightarrow x = \frac{4R}{3}$$

Now, to find $v_{\min}$, take the velocity at S to be zero

So, conserving energy, $GPE_i + KE_i = GPE_f + KE_f$

$$\left(-\frac{GMm}{R} - \frac{G(4M)m}{3R} \right) + \frac{1}{2}mv_{\min}^2 = \left(-\frac{GMm}{\left(\frac{4R}{3}\right)} - \frac{G(4M)m}{\left(\frac{8R}{3}\right)} \right) + 0 \Rightarrow v_{\min} = \sqrt{\frac{GM}{6R}}$$

Now, let the velocity with which the particle reaches the surface of the second planet be v_2

So, conserving energy, $GPE_i + KE_i = GPE_f + KE_f$

$$\left(-\frac{GMm}{R} - \frac{G(4M)m}{3R} \right) + \frac{1}{2}m\left(\sqrt{\frac{GM}{6R}}\right)^2 = \left(-\frac{GMm}{3R} - \frac{G(4M)m}{R} \right) + \frac{1}{2}mv_2^2 \Rightarrow v_2 = \sqrt{\frac{25GM}{6R}}$$

Now, let the velocity with which the particle reaches the mid-point of AB be v_M

So, conserving energy, $GPE_i + KE_i = GPE_f + KE_f$

$$\left(-\frac{GMm}{R} - \frac{G(4M)m}{3R} \right) + \frac{1}{2}m\left(\sqrt{\frac{GM}{6R}}\right)^2 = \left(-\frac{GMm}{2R} - \frac{G(4M)m}{2R} \right) + \frac{1}{2}mv_M^2 \Rightarrow v_M = \sqrt{\frac{GM}{2R}}$$

63. [A – R] [B – S] [C – T] [D – Q]

Let the centre of the disc be O. Now, the distance between the centre of the rod and the centre of the

$$\text{disc, } OP = \frac{R}{2}$$

At $t = t_1$, let the velocity of the centre of the disc be v_1

As the disc is rolling without slipping, the angular velocity of the disc, $\omega_1 = \frac{v_1}{R}$

To find the velocity of the centre of the rod, v_{P1} , we must treat it as a point on the disc itself. So, its velocity is the vector sum of two parts – one due to translation and the other due to rotation.

$$\text{So, } v_{P1} = \sqrt{v_1^2 + \left(\frac{v_1}{2}\right)^2} = \frac{\sqrt{5}v_1}{2}$$

Also, the angular velocity of the rod is the same as the angular velocity of the disc.

Now, conserving energy between $t = 0$ and $t = t_1$,

$$mg\left(\frac{R}{2}\right) = \frac{1}{2}mv_{P1}^2 + \frac{1}{2}\left(\frac{1}{12}m(\sqrt{3}R)^2\right)\omega_1^2$$

$$\Rightarrow \frac{mgR}{2} = \frac{1}{2}m\left(\frac{\sqrt{5}v_1}{2}\right)^2 + \frac{1}{8}mR^2\left(\frac{v_1}{R}\right)^2 \Rightarrow v_1 = \sqrt{\frac{2gR}{3}}$$

Therefore, the velocity of the centre of the disc, $v_{P1} = \frac{\sqrt{5}v_1}{2} = \sqrt{\frac{5gR}{6}}$

Now, at $t = t_2$, let the velocity of the centre of the disc be v_2

As the disc is rolling without slipping, the angular velocity of the disc, $\omega_2 = \frac{v_2}{R}$

To find the velocity of the centre of the rod, v_{P1} , we must treat it as a point on the disc itself. So, its velocity is the vector sum of two parts – one due to translation and the other due to rotation.

To find the velocity of the centre of the rod, v_{P2} , we must treat it as a point on the disc itself. So, its velocity is the vector sum of two parts – one due to translation and the other due to rotation.

So, $v_{P2} = v_2 - \left(\frac{R}{2}\right)\omega_2 = v_2 - \frac{v_2}{2} = \frac{v_2}{2}$

Also, the angular velocity of the rod is the same as the angular velocity of the disc.

Now, conserving energy between $t = 0$ and $t = t_2$,

$$mgR = \frac{1}{2}mv_{P2}^2 + \frac{1}{2}\left(\frac{1}{12}m(\sqrt{3}R)^2\right)\omega_2^2$$

$$\Rightarrow mgR = \frac{1}{2}m\left(\frac{v_1}{2}\right)^2 + \frac{1}{8}mR^2\left(\frac{v_2}{R}\right)^2 \Rightarrow v_2 = \sqrt{4gR}$$

Therefore, the velocity of the centre of the disc, $v_{P2} = \frac{v_1}{2} = \sqrt{gR}$

64. [A – R] [B – U] [C – P]

Let the velocity of the centre of the rod be $\vec{v}_C$

We know that the velocity of any point on the rod is a vector sum of two parts – one due translation, i.e. $\vec{v}_C$, and the other due to rotation, i.e. $r\omega$ directed appropriately. Here, r is the distance of the particular point from the centre of mass of the rod.

Now, let the part of velocity due to rotation for the end-points of the rod be called $\vec{v}_{RA}$ and $\vec{v}_{RB}$.

We know that $|\vec{v}_{RA}| = |\vec{v}_{RB}| = \frac{L}{2}\omega$

Furthermore, the directions of $\vec{v}_{RA}$ and $\vec{v}_{RB}$ are opposite to each other.

Therefore, $\vec{v}_{RB} = -\vec{v}_{RA}$

We already know that $\vec{v}_A = \vec{v}_C + \vec{v}_{RA}$ and $\vec{v}_B = \vec{v}_C + \vec{v}_{RB}$

Adding the two equations above, we get

$$\vec{v}_A + \vec{v}_B = 2\vec{v}_C + \vec{v}_{RA} + \vec{v}_{RB} = 2\vec{v}_C \Rightarrow \vec{v}_C = \frac{\vec{v}_A + \vec{v}_B}{2}$$

Using this in the equations above, we get

$$\vec{v}_{RA} = \vec{v}_A - \vec{v}_C = \vec{v}_A - \frac{\vec{v}_A + \vec{v}_B}{2} = \frac{\vec{v}_A - \vec{v}_B}{2}$$

Now since $|\vec{v}_{RA}| = \frac{L}{2}\omega$, $\omega = \frac{|\vec{v}_A - \vec{v}_B|}{L}$

Also, since the components of $\vec{v}_A$ and $\vec{v}_B$ along the length of the rod must be equal (because the rod is of constant length), $\vec{v}_A - \vec{v}_B$ is perpendicular to the rod.

65. [A – Q] [B – P] [C – U] [D – S] [E – R] [F – T]

(QPUSRT)

Let us think of the velocity of the particle in two components – one component along the axis of the helix (henceforth denoted by $v_{\parallel}$) and the other component in a plane perpendicular to the axis (denoted by $v_{\perp}$). The component of velocity along the axis of the helix remains constant. And the other component will rotate in a plane perpendicular to the axis. At $t=0$, $v_{\parallel} = v_0 \cos \theta$, and it will remain constant. Since $\sin \theta \hat{i} + \cos \theta \hat{j}$ is a unit vector along the axis, $v_{\parallel}$ can be written in vector form as $\vec{v}_{\parallel} = v_0 \cos \theta (\sin \theta \hat{i} + \cos \theta \hat{j})$

At $t=0$, the component of velocity in the plane perpendicular to the plane of the axis, i.e. $v_{\perp} = v_0 \sin \theta$, and it is directed along the unit vector $(-\cos \theta \hat{i} + \sin \theta \hat{j})$. This component will rotate in this plane such it will oscillate with time period T between two directions, which are $(-\cos \theta \hat{i} + \sin \theta \hat{j})$ and $\hat{k}$. This means that

At $t=0$, $v_{\perp}$ is entirely along the direction $(-\cos \theta \hat{i} + \sin \theta \hat{j})$

At $t = \frac{T}{4}$, $v_{\perp}$ is entirely along the direction $\hat{k}$

At $t = \frac{T}{2}$, $v_{\perp}$ is entirely along the direction $(\cos \theta \hat{i} - \sin \theta \hat{j})$

At $t = \frac{3T}{4}$, $v_{\perp}$ is entirely along the direction $(-\hat{k})$... and so on.

But the magnitude of $v_{\perp}$ will remain $v_0 \sin \theta$. Therefore, we can write $v_{\perp}$ in vector form as

$$\vec{v}_{\perp} = [v_0 \sin \theta] \left[(-\cos \theta \hat{i} + \sin \theta \hat{j}) \cos \left(\frac{2\pi}{T} t \right) + (\hat{k}) \sin \left(\frac{2\pi}{T} t \right) \right]$$

So, the instantaneous velocity of the particle,

$$\begin{aligned} \vec{v} = \vec{v}_{\parallel} + \vec{v}_{\perp} &= \left[v_0 \sin \theta \cos \theta \left(1 - \cos \left(\frac{2\pi}{T} t \right) \right) \right] \hat{i} \\ &+ \left[v_0 \left(\cos^2 \theta + \sin^2 \theta \cos \left(\frac{2\pi}{T} t \right) \right) \right] \hat{j} + \left[v_0 \sin \theta \sin \left(\frac{2\pi}{T} t \right) \right] \hat{k} \end{aligned}$$

Now, since at $t=0$ the particle is at the origin, the instantaneous position of the particle is

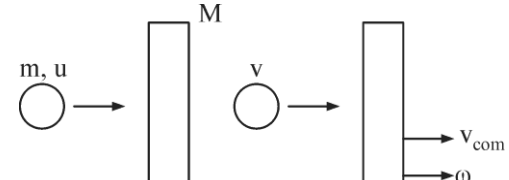
$$\vec{r} = \int_0^t \vec{v} dt$$

$$\begin{aligned} \text{Therefore, } \vec{r} &= \left[v_0 \sin \theta \cos \theta \left(t - \left(\frac{T}{2\pi} \right) \sin \left(\frac{2\pi}{T} t \right) \right) \right] \hat{i} \\ &+ \left[v_0 \left(\left(\cos^2 \theta \right) t + \left(\frac{T}{2\pi} \sin^2 \theta \right) \sin \left(\frac{2\pi}{T} t \right) \right) \right] \hat{j} \\ &+ \left[(v_0 \sin \theta) \left(\frac{T}{2\pi} \right) \left(1 - \cos \left(\frac{2\pi}{T} t \right) \right) \right] \hat{k} \end{aligned}$$

SECTION 5

NUMERICAL TYPE

$$66. (2/3) mu = mv + VM_{com} ; \quad mud = mvd + \frac{ML^2}{3} \omega ; \quad V_{com} = \omega L / 2$$

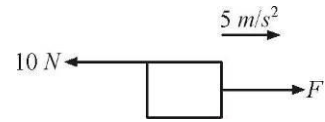
$$mud = mvd + \frac{ML^2}{V} \cdot \frac{V_{com}}{L} \therefore \text{comparing eqns. } d = \frac{2L}{3}$$


71.(40) Limiting friction between the blocks

$$f_L = 10N; \quad [a_{\max}]_A = \frac{10}{2} = 5 \text{ m/s}^2$$

i.e., for slipping between A and B, B must move with 5 m/s^2

$$\Rightarrow F = 25N, \quad \text{But given } F = 20N$$



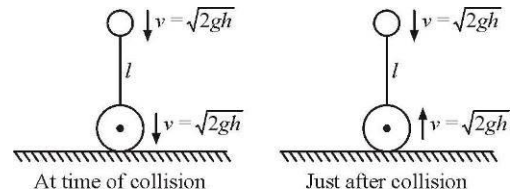
72.(1) Just after collision velocities of both the balls will be $\sqrt{2gh}$ in opposite direction. Relative acceleration between the two balls is zero and relative velocity of approach is $2\sqrt{2gh}$. Hence they

$$\text{will collide after a time } t = \frac{l}{2\sqrt{2gh}}$$

At the time of collision of two balls relative velocity of approach = relative velocity of separate = $2\sqrt{2gh}$

Hence string becomes tight after the same time

$$t = \frac{l}{2\sqrt{2gh}}. \quad \text{Hence the total time will be } (2t) = \frac{l}{\sqrt{2gh}} = 1s$$



73.(0.85) Conservation of angular momentum about 'O'

$$mv_x R = (M + m)v_{\text{com}} \times \frac{R}{2} + I_{\text{com}}\omega$$

$$\omega = 0.85 \text{ radian}$$

$$74.(4) \quad \frac{Mg}{2} + T_3 - (T_1 + T_2) = \frac{Ma}{2} \quad \dots(i)$$

$$Mg - T_3 = Ma \quad \dots(ii)$$

From (i) and (ii)

$$\frac{3}{2}Mg - (T_1 + T_2) = \frac{3Ma}{2}; \quad T_1 \times R = \frac{MR^2}{2} \alpha_1$$

$$\frac{2T_1}{MR} = \alpha_1; \quad \alpha_1 R = 2a$$

$$T_1 = Ma; \quad (T_2 - T_1) \times \frac{R}{2} = \frac{M}{2} \times \left(\frac{R}{2}\right)^2 \times \frac{1}{2} \times \alpha_2$$

$$(T_2 - T_1) = MR\alpha_2; \quad \alpha_2 R = 2a$$

$$8(T_2 - T_1) = 2Ma; \quad 8T_2 = 10Ma$$

$$T_2 = \frac{5}{4}Ma; \quad \frac{3Mg}{2} - \frac{9}{4}Ma = \frac{3Ma}{2}; \quad \frac{3Mg}{2} = \frac{15Ma}{4}; \quad a = \frac{2g}{5} = \frac{xg}{5} = 4$$

$$75.(4) \quad 2mg - f - T_1 = 4ma_{\text{com}} \quad \dots(i)$$

$$\alpha R + a_{\text{com}} - a = 0 \quad \dots(ii)$$

$$I\alpha = (T_1 - f)R \quad \dots(iii)$$

$$2mg + f - T_2 = 4ma \quad \dots(iv)$$

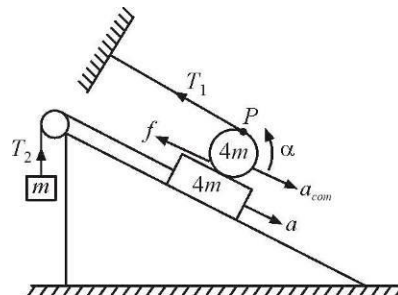
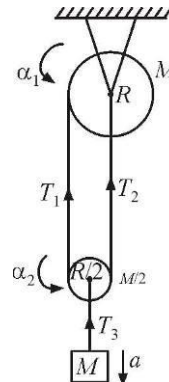
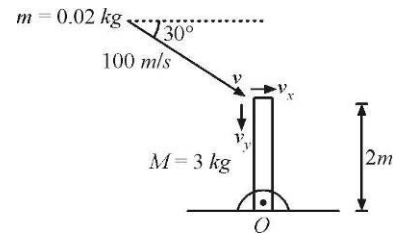
$$T_2 - mg = ma \quad \dots(v)$$

$$\alpha R = a_{\text{com}} \quad (\text{Pure rolling at } P)$$

$$\alpha \frac{4mR^2}{2} = (T_1 - f)R; \quad a_{\text{com}} = \alpha R = \frac{(T_1 - f)}{2m}$$

$$mg + f = 5ma \quad [\text{From (iv) and (v)}]$$

$$\frac{T_1 - f}{m} - a = 0; \quad \frac{T_1 - f}{m} - \frac{mg - f}{5m} = 0$$



$$5T_1 - 6f - mg = 0$$

$$5T_1 - 6f = mg \quad \dots(A)$$

$$10ma_{com} - f = mg \quad \dots(B)$$

$$10mg - 5f - 5T_1 - 2f = 2mg$$

$$8mg = 7f + 5T_1 \quad \dots(C)$$

$$\text{From (A) and (C) } -13f = -7mg; \quad f = \frac{7}{13}mg$$

Put f in (B)

$$10ma_{com} - \frac{7}{13}mg = mg; \quad a_{com} = \frac{2g}{13} = \alpha R; \quad a = \alpha R + a_{com} = \frac{2g}{13R} \times R + \frac{2g}{13} = \frac{4g}{13}; \quad k = 4$$

76.(5) $\omega R = v$

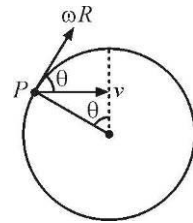
For pure rolling vertical component of velocity of P is $\omega R \sin \theta = v \sin \theta$

$$\text{Maximum height attended} = \frac{v^2 \sin^2 \theta}{2g}$$

$$\text{Total height from ground } H = R(1 + \cos \theta) + \frac{v^2 \sin^2 \theta}{2g}$$

$$\frac{dH}{d\theta} = -R \sin \theta + \frac{v^2}{2g} 2 \sin \theta \cos \theta \quad \frac{dH}{d\theta} = 0 \quad \Rightarrow \quad \frac{v^2 \cos \theta}{g} = R$$

$$= \frac{10 \times 10 \cos \theta}{10} = 2; \quad \cos \theta = \frac{1}{5}; \quad \sec \theta = 5$$

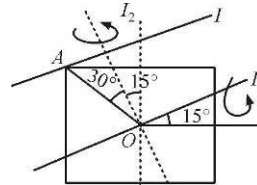


77.(4) By symmetry, $I_1 = I_2$, but $I_1 + I_2 = \frac{ML^2}{6}$

$$\Rightarrow \quad I_1 = \frac{ML^2}{12}$$

$$I = I_1 + M(AO \cos 30^\circ)^2$$

$$I = \frac{ML^2}{12} + M \left(\frac{L}{\sqrt{2}} \times \frac{\sqrt{3}}{2} \right)^2 = \frac{11}{24} ML^2 \quad \Rightarrow \quad I = \frac{11}{24} \times \frac{4}{11} \times 4 = 4$$



78.(61) Velocity at the end of $(n-1)$ seconds,

$$v_{n-1} = a_0(1) + \frac{a_0}{2}(1) + \frac{a_0}{4}(1) + \dots + \frac{a_0}{2^{n-1}}(1) = \frac{a_0(2^{n-1} - 1)}{2^{n-2}}$$

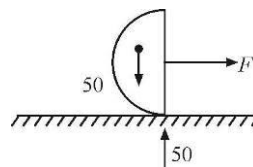
So, displacement during the n^{th} second: $D_n = v_{n-1}(1) + \frac{1}{2} \left(\frac{a_0}{2^{n-1}} \right) (1)^2 = \frac{a_0(2^{n+1} - 3)}{2^n}$

Therefore, $D_5 = \frac{61a_0}{32}$

79.(5) $F = \left(\frac{10}{2} \right) (10)^2 \times \frac{3 \times 0.8}{8} = F = m\omega^2 \left(\frac{3R}{8} \right)$

$$\omega = \frac{v}{R} = 10$$

$$= 5 \times 100 \times 0.3 = 150 \text{ N}$$



80.(4) By energy conservation between the lowest and the highest points in the circle,

$$\frac{1}{2}mv_{\max}^2 - \frac{1}{2}mv_{\min}^2 = mg(2L)$$

$$\Rightarrow \quad v_{\max}^2 - v_{\min}^2 = 4gL$$

And we are given that $\frac{v_{\max}}{v_{\min}} = \sqrt{2}$

Solving, we get $v_{\max} = \sqrt{8gL}$ and $v_{\min} = \sqrt{4gL}$

81.(2) let a_1 and a_2 be the tangential accelerations of two beads respectively. Then for bead A

$$m_1 g \sin 30^\circ + T \sin 45^\circ = m_1 a_1 \quad \dots(i)$$

For bead B

$$m_2 g \sin 60^\circ - T \sin 45^\circ = m_2 a_2 \quad \dots(ii)$$

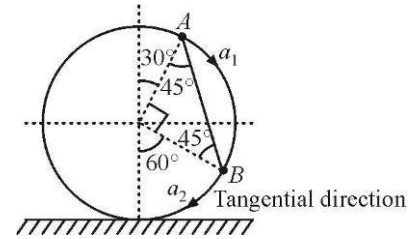
Since string remains taut $\therefore a_1 \cos 45^\circ = a_2 \cos 45^\circ$

(Acceleration along string remains same)

$$\Rightarrow a_1 = a_2 = a \quad \therefore \text{From (i) and (ii)}$$

$$a = \frac{g(m_1 + \sqrt{3}m_2)}{2(m_1 + m_2)}; \quad \frac{T}{\sqrt{2}} = \frac{m_1 g(m_1 + \sqrt{3}m_2)}{2(m_1 + m_2)} - \frac{m_1 g}{2}$$

$$T = \frac{(\sqrt{3}-1)(m_1 m_2 g)}{\sqrt{2}(m_1 + m_2)}$$



82.(7) $\mu_s g = \frac{dv}{dt}$ at $t = 0$ sec $\therefore \mu_s = 0.4$

At $t = 3$ sec velocities of plank and block are equal.

$$\therefore 2 + \mu_k g(2) = v \text{ at } t = 3 \text{ sec} = 8 \quad \therefore \mu_k = 0.3 \quad \therefore s + k = 4 + 3 = 7$$

83.(3) Since $\frac{\mu}{4} < \frac{\mu}{2} < \mu$, the upper block will move faster than the middle block and hence force of friction on upper block is towards left

$$\therefore f = \frac{\mu}{4} mg \quad \therefore a_A = \frac{\mu g}{4} \text{ leftward}$$

$$f' = \frac{\mu}{2} \left(m + \frac{m}{2} \right) g = \frac{3}{4} \mu mg$$

$$\therefore a_B = \mu g \quad (\text{Leftward}) \quad \therefore a_{A/B} = \frac{3}{4} \mu g \quad (\text{Rightward})$$

For sliding down the block B, A has to move a distance $\frac{3L}{8}$ relative to block B.

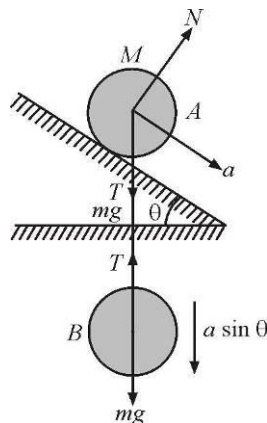
$$\frac{3L}{8} = \frac{1}{2} \frac{3}{4} \mu g t^2 \quad \therefore t = \sqrt{\frac{L}{\mu g}} = 3 \text{ sec}$$

84.(1) $Mg \sin \theta + T \sin \theta = Ma$

$$mg - T = m a \sin \theta$$

Solving, we get

$$a = \frac{(M+m)g \sin \theta}{M + m \sin^2 \theta}$$



5.(12) $-a_C + 2a_B + 2a_A = 0$

$$a_C = 2(a_B + a_A) \quad \dots(i)$$

$$-a_A - a_B + 2a_D = 0$$

$$2a_D = a_A + a_B \quad \dots(ii)$$

$$a_C + a_B = 6 \quad \dots(iii)$$

$$a_D - a_A = 11 \quad \dots (iv)$$

From (i) and (ii)

$$a_C = 2 \times 2a_D; \quad a_C = 4a_D$$

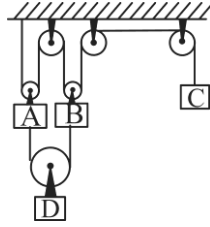
From (iii) and (iv)

$$a_C - a_D + a_A + a_B = -5$$

$$a_C + a_D = 5; \quad 5a_D = 5$$

$$a_D = 1; \quad a_C = 4$$

$$= 0 + at; \quad v = 4(3) = 12 \text{ m/s}$$



SECTION 6

SUBJECTIVE TYPE

86. (i) $\frac{12v}{7L}$; (ii) 3.5

(i) $mv \times \frac{L}{4} = \left[\frac{mL^2}{12} + m \left(\frac{L}{4} \right)^2 \right] \omega$; $w = \frac{12v}{7L}$

(ii) $\tau = \frac{dL}{dt} = \frac{dI}{dt} \omega = \omega \frac{d}{dt} (I_{rod} + I_{insect})$

$$= \omega \left[\frac{dI_{insect}}{dt} \right]$$

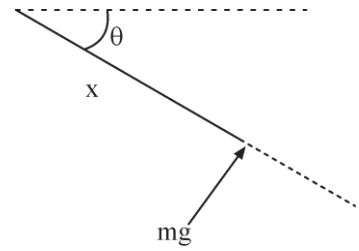
$$= \omega \frac{d(Mx^2)}{dt} = M\omega 2x \frac{dx}{dt} = 2M\omega x \frac{dx}{dt} = Mg x \cos \theta$$

$$\frac{2\omega}{g} dx = \cos \theta dt; \quad \frac{2\omega}{g} \int_{L/4}^{L/2} dx = \int_0^t \cos(\omega t) dt$$

$$\frac{2\omega}{g} (L/4) = \frac{1}{\omega} \sin \omega t; \quad \frac{2\omega}{g} \cdot \frac{L}{4} = \frac{\sin \omega}{\omega} (\pi / 2\omega)$$

$$\frac{\omega L}{2g} = \frac{1}{\omega}; \quad \omega^2 = \frac{2g}{L}; \quad \omega = \sqrt{\frac{2g}{L}}; \quad \frac{12v}{7L} = \sqrt{\frac{2g}{L}}; \quad v = \frac{7}{12} \sqrt{2gL}$$

$$= \frac{7}{12} \times \sqrt{2 \times 10 \times 1.8} = \frac{7}{12} \times 6 = 3.5 \text{ m/s}$$



87. (i) $\frac{Ft}{m_0 + rt}$; (ii) $\frac{m_0 F}{(m_0 + rt)^2}$

Force = F, mass of car = m_0

at any time t

mass = $m_0 + rt$

at time

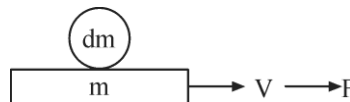
$$F_{net} = F - v \frac{dm}{dt}; \quad ma = F - v \frac{dm}{dt}$$

$$ma = F - v(\mu); \quad (m_0 + rt)a = F - rv$$

$$a = \frac{F - rv}{m_0 + rt}; \quad \frac{dv}{F - rv} = \frac{dt}{m_0 + rt}$$

$$-\frac{1}{r} \ln(F - rv) \Big|_0^v = \frac{1}{r} \ln(m_0 + rt) \Big|_0^t; \quad \ln \frac{F}{F - rv} = \ln \left(\frac{m_0 + rt}{m_0} \right)$$

$$\frac{F}{F - rv} = \frac{m_0 + rt}{m_0}; \quad \frac{m_0 F}{m_0 + rt} = F - rv; \quad v = \frac{Ft}{m_0 + rt}; \quad a = \frac{m_0 F}{(m_0 + rt)^2}$$



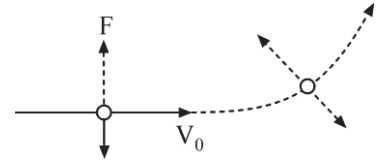
88. $\left\{ \frac{u}{v_0} \ln(m_0 / m) \right\}$

$F = \text{constant and } \perp \text{ to velocity } \therefore \text{ no change in velocity}$

$$F = -\frac{dm}{dt}u = \frac{mv_0^2}{R}; \quad -\frac{dm}{dt}u = \frac{mv_0^2}{R}; \quad -\int \frac{dm}{m} = \frac{v_0^2}{Ru} \int dt$$

$$-\ln m / m_0 = \frac{v_0^2}{Ru} t; \quad \frac{u}{v_0^2} \ln m_0 / m = \frac{1}{R}; \quad R = \frac{v_0^2 t}{u \ln(m_0 / m)}$$

$$v_0 t = R\theta; \quad v_0 t = \frac{v_0^2 t}{u \ln(m_0 / m)} \theta; \quad \theta = \frac{u}{v_0} \ln(m_0 / m)$$



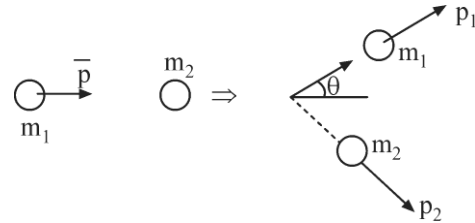
89. $\{ \sin \theta \leq m_2 / m_1 \}$

$$\bar{p} = \bar{p}_1 + \bar{p}_2; \quad p^2 + p_1^2 - 2pp_1 \cos \theta = p_2^2$$

$$\frac{p^2}{2m_1} = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2}; \quad \frac{p^2}{2m_1} = \frac{p_1^2}{2m_1} + \frac{p^2 + p_1^2 - 2p_1 p_2 \cos \theta}{2m_2}$$

$$\frac{p^2}{2} \left(\frac{1}{m_1} - \frac{1}{m_2} \right) = \frac{p_1^2}{2} \left(\frac{1}{m_2} + \frac{1}{m_1} \right) - \frac{2p_1 p_2 \cos \theta}{2m_2}$$

$$P^2 \left(\frac{m_2 - m_1}{m_1 m_2} \right) = P_1^2 \left(\frac{m_1 + m_2}{m_1 m_2} \right) - 2P \frac{P_1 \cos \theta}{m_2}$$



Quadratic in p

$$D \geq 0; \quad \frac{(2p_1 \cos \theta)^2}{(m_2)^2} + 4 \left(\frac{m_2 - m_1}{m_1 m_2} \right) (P_1^2) \left(\frac{m_1 + m_2}{m_1 m_2} \right) \geq 0$$

$$\cos^2 + \frac{m_2^2 - m_1^2}{m_1^2} \geq 0; \quad \left(\frac{m_2}{m_1} \right)^2 \geq \sin^2 \theta; \quad \sin \theta \leq m_2 / m_1$$

90. $\frac{2m_1}{m_1 + m_2}$

$$v \cos \theta = u \sin \theta; \quad v = u \tan \theta$$

$$m_1 u = m_2 v_2 \cos \theta; \quad v_2 = \frac{m_1}{m_2} \frac{u}{\cos \theta}$$

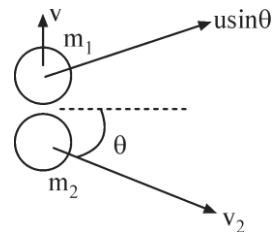
$$\frac{1}{2} m_1 u^2 = \frac{1}{2} m_1 v^2 + \frac{1}{2} m_2 \frac{m_1^2}{m_2^2} \frac{u^2}{\cos^2 \theta}$$

$$u^2 = u^2 \tan^2 \theta + \frac{m_1}{m_2 \cos^2 \theta}; \quad \cos^2 \theta - \sin^2 \theta = m_1 / m_2$$

$$\cos^2 \theta + \sin^2 \theta = 1; \quad 2 \cos^2 \theta = \frac{m_1 + m_2}{m_2}$$

$$\cos^2 \theta = \frac{m_1 + m_2}{2m_2}; \quad \sin^2 \theta = \frac{m_2 - m_1}{2m_2}$$

$$\tan^2 \theta = \frac{m_2 - m_1}{m_1 + m_2}; \quad \frac{\Delta KE}{KE} = 1 - \tan^2 \theta = \frac{2m_1}{m_1 + m_2}$$



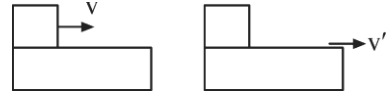
91. $-\frac{mM}{M+m} gh$; $v = \sqrt{2gh}$

$$m(\sqrt{2gh}) = (M+m)v^1; \quad v^1 = \frac{m}{m+M} \sqrt{2gh}; \quad w_f = \Delta k \epsilon$$

$$= \frac{1}{2}(M+m) \left(\frac{m}{M+m} \right)^2 (2gh) - \frac{1}{2}(2gh)$$

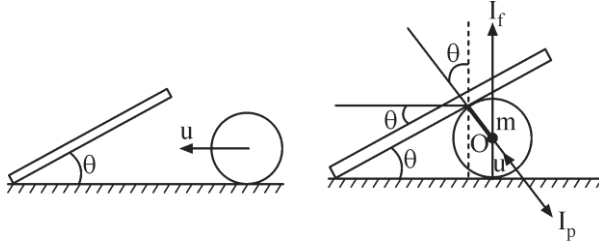
$$= \frac{1}{2} \frac{m^2}{(M+m)} (2gh) - \frac{1}{2} m(2gh) = \frac{1}{2} m(2gh) \left(\frac{m}{M+m} - 1 \right)$$

$$= \frac{1}{2} m(2gh) \left(\frac{-m}{M+m} \right) = \frac{mM}{M+m} - gh$$



92. (i) $\frac{2mu}{\sin \theta}$ (ii) $2mu \cot \theta$

$$I_p \sin \theta = 2mu; \quad I_p = \frac{2mu}{\sin \theta}$$



$$I_p \cos \theta = I_f; \quad I_f = 2mu \cot \theta$$

93. $\left\{ \frac{L \cos \theta \sin^2 \theta}{2h - L \cos \theta \sin \theta} \right\}$

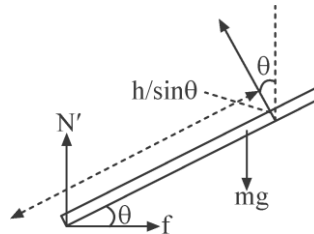
$$N \cos \theta + N' = mg; \quad N \sin \theta = f = \mu N'; \quad \frac{Nh}{\sin \theta} = mg \left(\frac{L}{2} \cos \theta \right)$$

$$(mg - N')h / \sin \theta = mg \frac{L}{2} \cos \theta$$

$$\frac{mg h / \sin \theta - mg L / 2 \cos \theta}{h / \sin \theta} = N'$$

$$\mu \left(h / \sin \theta - \frac{L}{2} \cos \theta \right) = \left(\frac{L}{2} \cos \theta \right) \sin \theta$$

$$\mu = \frac{L \cos \theta \sin^2 \theta}{2h - L \cos \theta \sin \theta}$$



94. $\frac{\omega^2}{\frac{8\pi}{R} [\mu g] \frac{(1+\mu)}{1+\mu^2}}$

$$N_2 = mg - F_2;$$

$$N_1 = f_1$$

$$f_1 = \mu N_2;$$

$$f_2 = \mu N_1$$

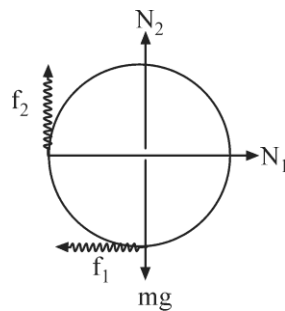
$$N_2 = mg - \mu N_1;$$

$$N_1 = \mu N_2$$

$$N_2 = mg - (\mu N_2)$$

$$N_2 = \frac{mg}{1+\mu^2}; \quad N_1 = \frac{\mu mg}{1+\mu^2}; \quad f_1 = \frac{\mu mg}{1+\mu^2}; \quad f_2 = \frac{\mu^2 mg}{1+\mu^2}$$

$$\alpha = \frac{\left(\frac{\mu mg}{1+\mu^2} + \frac{\mu^2 mg}{1+\mu^2} \right) R}{mR^2 / 2}; \quad \omega^2 + \omega_0^2 = 2 \left(\frac{2}{R} [\mu g] \left[\frac{1+\mu}{1+\mu^2} \right] \theta \right);$$



$$n = \frac{\theta}{2\pi}; \quad \omega^2 = \frac{8\pi}{R} [\mu g] \frac{(1+\mu)}{1+\mu^2}$$

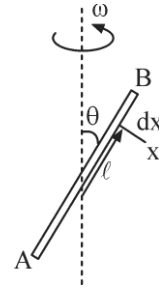
95. $\frac{m\omega^2}{12} \sin \theta \cos \theta L^2$; $dm = \frac{m}{\ell} dx$; $r = x \sin \theta$

$$dF = dm\omega^2 (x \sin \theta) \quad d\tau = dF \times x \cos \theta$$

$$\tau = \frac{m}{\ell} \omega^2 \sin \theta \int x dx \cos \theta$$

$$= \frac{m\omega^2}{\ell} \sin \theta \cos \theta \left[\frac{x^3}{3} \right]_0^{\ell/2} \times 2$$

$$= \frac{2m\omega^2}{\ell} \frac{\sin \theta \cos \theta}{3} \cdot \frac{\ell^3}{8} = \frac{m\omega^2}{12} \sin \theta \cos \theta L^2$$



96. $\frac{3mg}{4} \sin \theta [3 \cos \theta - 2]$

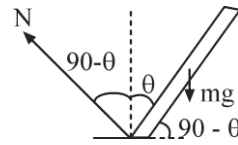
$$mg \left(\ell/2 - \frac{\ell}{2} \cos \theta \right) = \frac{1}{2} \frac{m\ell^2}{3} \omega^2$$

$$\omega^2 = \frac{3g}{\ell} (1 - \cos \theta); \quad \frac{\ell}{2} mg \sin \theta = \frac{m\ell^2}{3} \alpha$$

$$\alpha = \frac{3g \sin \theta}{\ell^2} \cdot \frac{\ell}{2} = \frac{3g \sin \theta}{2\ell}$$

$$a_t = \frac{3g \sin \theta}{2\ell} \times \frac{\ell}{2} \Rightarrow \frac{m3g \sin \theta}{4} = mg \sin \theta - L$$

$$a_t = \frac{3}{4} g \sin \theta; \quad a_n = \frac{\omega^2 \ell}{2} = \frac{3g}{2} (1 - \cos \theta)$$



$$\begin{aligned} \text{Total acc. of rod's centre} &= \sqrt{\left(\frac{3}{4} g \sin \theta \right)^2 + \left(\frac{3g}{2} (1 - \cos \theta) \right)^2} \\ &= \frac{3g}{4} [1 - \cos^2 \theta + 4 \cos^2 \theta - 8 \cos \theta]^{1/2} \\ &= \frac{3g}{4} [5 + 3 \cos^2 \theta - 8 \cos \theta]^{1/2} \end{aligned}$$

$$mg - N = ma_y = m(a_N \cos \theta + a_T \sin \theta); \quad mg - \frac{3mg}{4} (2 \cos \theta - 3 \cos^2 \theta + 1)$$

$$f = m(a_t \cos \theta - a_N \sin \theta) = m \left(\frac{3}{4} g \sin \theta \cos \theta - \frac{3g}{2} (\sin \theta - \cos \theta \sin \theta) \right)$$

$$= \frac{3mg}{4} [\sin \theta \cos \theta - 2 \sin \theta + 2 \cos \theta \sin \theta] = \frac{3mg}{4} \sin \theta [3 \cos \theta - 2]$$

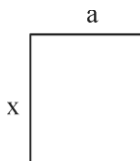
97. $\left\{ \frac{2m Ma}{M^2 - m^2} \right\}$

$$mgx = M(\sqrt{a^2 + x^2} - a)g$$

$$mx + Ma = M\sqrt{abx^2}$$

$$m^2 x^2 + M^2 a^2 + 2mMax = M^2 a^2 + M^2 x^2$$

$$(M^2 - m^2)x^2 = 2mMax; \quad x = \frac{2m Ma}{M^2 - m^2}$$



98. $\frac{\lambda}{2}v^3 \rightarrow dl, dt;$

$$\frac{d}{dx} \\ d\ell = 2dx$$

In time dt if A moves dl , $d\ell = 2dx$ mass λdx attains v .

$$dp = (\lambda dx)v; \quad F = \frac{dp}{dt} = \lambda v \frac{dx}{dt} =$$

$$v = \frac{dl}{dt} = \frac{2dx}{dt}; \quad \frac{\lambda}{2}v^2 = \frac{\lambda}{2}v^2 \times v = \frac{\lambda}{2}v^3$$

99. $-\frac{1}{2}v_0^2 m \sqrt{gb}$

$$\mu = bx; \quad f = -bx \, mg$$

$$a = -g \, bx; \quad \int v \, dv = -\int gbx \, dx$$

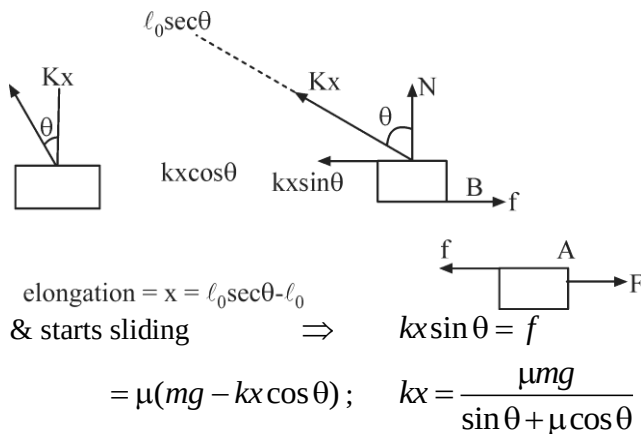
$$\frac{-v_0^2 + v^2}{2} = -gb \frac{x^2}{2}; \quad v = \sqrt{v_0^2 - gbx^2}$$

$$P = -mgb \sqrt{v_0^2 - gbx^2}; \quad \frac{dp}{dx} = 0$$

$$v_0^2 - gbx^2 - gbx^2 = 0; \quad gbx^2 = \frac{v_0^2}{2}$$

$$x = \frac{v_0}{\sqrt{2}\sqrt{gb}}; \quad P = -mgb \frac{v_0}{\sqrt{2}\sqrt{gb}} \cdot \sqrt{\frac{v_0^2}{2}} = -\frac{1}{2}v_0^2 m \sqrt{gb}$$

100. $\frac{1}{2} \left(\frac{\mu mg}{\sin \theta + \mu \cos \theta} \right) l_0 (\sec \theta - 1)$



Till this position

$$\text{Work done} = \frac{1}{2} kx^2 = \frac{1}{2} \left(\frac{\mu mg}{\sin \theta + \mu \cos \theta} \right) l_0 (\sec \theta - 1)$$